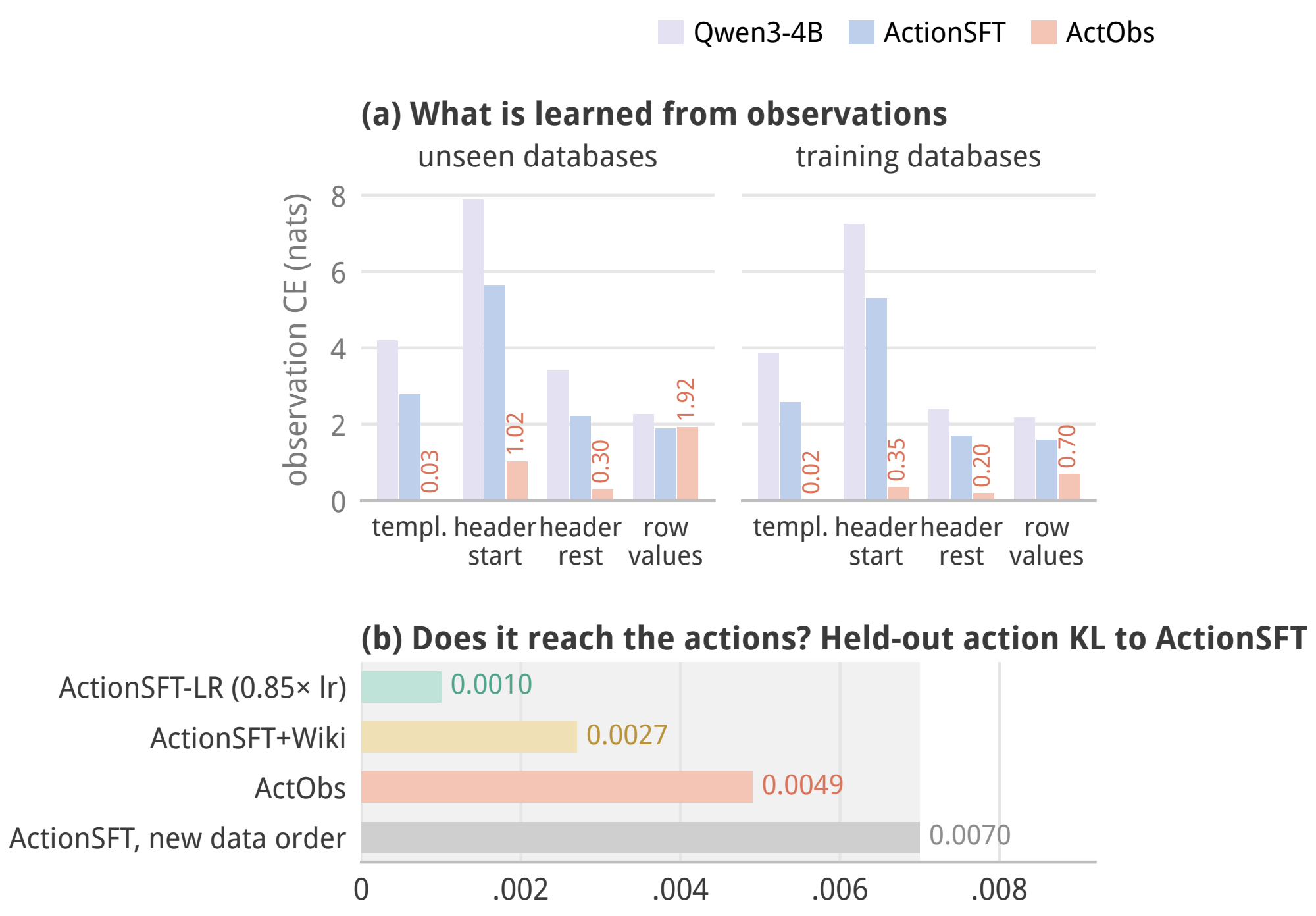
Qwen3-4B
ActionSFT
ActObs
(a) What is learned from observations
unseen databases
training databases
observation CE (nats)
0
2
4
6
8
0.03
1.02
0.30
1.92
0.02
0.35
0.20
0.70
templ.
header start
header rest
row values
templ.
header start
header rest
row values
(b) Does it reach the actions? Held-out action KL to ActionSFT
ActionSFT-LR (0.85× lr)
ActionSFT+Wiki
ActObs
ActionSFT, new data order
0.0010
0.0027
0.0049
0.0070
0
.002
.004
.006
.008